TEST PLAN

Item	Details
Project	Automation Exercise – Online Shopping Platform
Website URL	https://www.automationexercise.com/
Document	Test Plan
Team Leader	Mohamed Alhassan Ali

1. Project Overview

Automation Exercise is an e-commerce website used to test functionalities such as user registration, product browsing, cart management, and the checkout process. Testing will focus on verifying the core modules and ensuring system stability across different test types.

Purpose

- Verify that all user functionalities work as expected.
- Ensure data integrity across UI, API, and database layers.
- Identify UI, functional, and performance defects.
- Validate checkout, payment simulation, and order flows.

2. Test Objectives

The primary objective of this testing phase is to ensure that the Automation Exercise web application and its public APIs work reliably across the core user journeys, and to identify, log, and track defects before they affect users. Specific objectives:

1. **Functional and End-to-End Accuracy:** Verify the complete user workflows, including registration, login, logout, account deletion, product browsing and search, category/brand filtering, shopping cart, checkout, and payment with dummy data, with attention to invalid inputs and an already registered email.
2. **Backend API Verification:** Validate the public REST APIs (products list, brands list, search product, verify login, and create, update, get and delete user account) using Postman, checking responses and status codes.
3. **Test Automation:** Automate the main regression scenarios (login, cart, checkout) using Selenium WebDriver and TestNG, and run them on Chrome, Firefox, and Edge.
4. **Defect Management:** Log every defect in Jira with steps to reproduce and severity and track it until it is retested and closed.
5. **Quality Criteria:** Execute 100% of the planned test cases, reach a pass rate of at least 90%, and have no open Critical or High severity defects before the final report.

3. Scope

In Scope	Out of Scope
UI Testing (Manual + Automation): • Registration and login (valid and invalid cases, duplicate email) • Product search and filtering by category and brand • Adding and removing products in the Cart, and verifying prices and totals • Full Checkout journey through order confirmation • Contact Us form • Subscription	• Deep security testing (penetration testing) • Native mobile apps (web site only) • Real payments and payment gateways (test data only) • Full accessibility and WCAG compliance • Localization and multi-language support • Internal back-end or direct database testing • Very old browsers and uncommon devices • Third-party ads and external content

In Scope	Out of Scope
• Navigation, buttons, and key links • Cross-browser: Chrome and Firefox at minimum **API Testing (Postman + Newman):** • Get All Products / Get All Brands • Search Product (with and without the parameter) • Verify Login (valid, invalid, and missing credentials) • Create / Update / Delete Account • Get User Detail by Email • Validate status codes, payloads, and error messages **Performance Testing (JMeter):** • Load test on Home, Products, and Login pages • Stress test on the main API endpoints **Process:** • Exploratory testing • BDD with Cucumber for the core flows • Defect tracking in Jira • Continuous execution through Jenkins	

Interfaces & Workflows

- UI testing through the web interface.
- API interface through Postman.
- Workflow: Product → Cart → Checkout → Confirmation.
- Vendor Integration: none.

4. Testing Types

Type	Description
Manual Testing	Executing functional, positive, and negative test scenarios across all core user journeys (Registration, Login, Cart, and Checkout). Test cases are documented in Excel/Google Sheets, and defects are logged and tracked in Jira
Automation Testing	Automating core regression test scenarios (Login, Shopping Cart, and Checkout) using Selenium WebDriver with Java and TestNG, structured around the Page Object Model (POM) design pattern for maintainability
API Testing	Validating public REST API endpoints (product catalog, brands list, search product, authentication, and CRUD operations for user accounts) using Postman collections to check response bodies and HTTP status codes
Regression Testing	Re-executing automated test scripts and core manual test cases after bug fixes or environment updates to ensure existing functionalities remain unaffected
Compatibility Testing	Verifying the web application's UI, layouts, and functionalities across multiple web browsers (Google Chrome, Mozilla Firefox, and Microsoft Edge) to ensure uniform user experience
Usability & Exploratory Testing	Evaluating UI navigation, layout, and error messages, and conducting ad-hoc exploratory sessions to discover unscripted defects and usability issues

5. Testing Strategy

The testing strategy combines manual and automated testing. Manual testing is used first to understand the application and find functional and usability defects. Stable scenarios are then automated for regression. API

testing is performed in parallel on the application's public endpoints. Each phase is documented, and the results are reported in the Test Summary Report.

Responsibilities

Role	Responsibility
Team Leader	Define the test strategy, review the test plan and reports, track entry and exit criteria, coordinate the work between members, and approve the final deliverables
Testers	Write test cases, execute manual, automated, and API tests, report and retest defects in Jira, and log execution results
Mentor / Reviewer	Give feedback on the plan and the test results

Test Assumptions

- The test environment (https://www.automationexercise.com/) is accessible, stable, and available throughout execution.
- Test automation tools (Selenium, TestNG, Postman) and the defect tracking platform (Jira) are configured and accessible to all team members.
- Requirements and core workflows remain unchanged during the testing cycle.
- Dummy data is used for all accounts, addresses, and payment details.

Data Approach

- **Dynamic data generation:** unique emails and user profiles are generated during account creation scenarios (using timestamps or unique strings) to prevent duplication errors, and each automated test creates its own account, so tests do not depend on each other.
- **Payment data:** dummy card details are used to simulate checkout, and no real payment information is entered.
- **Static test sets:** predefined product keywords, categories, and review text are used for search, filtering, and API test collections.
- No real or confidential personal data is used.

6. Test Environment & Tools

The testing will be performed on the Automation Exercise website.

Environment Details

Component	Details
Website URL	https://www.automationexercise.com/
Operating System	Windows 10 / Windows 11
Browsers	Google Chrome, Microsoft Edge
Automation Tool	Selenium WebDriver
Programming Language	Java
Testing Framework	TestNG
IDE	IntelliJ IDEA / Eclipse
Test Data	Dummy user accounts, emails, and product information
Internet Connection	Stable internet connection
API Tool	Postman
Database	MySQL

Component	Details
Test Management	Jira
Test Documentation	Excel (for writing and tracking test cases and results)
Version Control	GitHub
Reporting	Allure, Extent Reports

Testing Setup

- Testing will be performed through the web browser.
- Selenium WebDriver will be used to automate test cases.
- Java will be used to write automation scripts.
- TestNG will be used to organize and execute test cases.
- Dummy data will be used during testing.

Test Data

Data Type	Example
User Account	Email: user@example.com – Password: uh412
Product	Blue Top, Men Tshirt, etc.
Quantity	1 to 10 items
Address	Example: Cairo, Egypt
Payment	Dummy test card details

7. Levels of Testing

Functional Testing – Features to be Tested

ID	Scenario	Priority
User Registration & Account		
TS_01	Register a new user with valid data	High
TS_02	Register with an already existing email	High
TS_03	Register with empty or invalid mandatory fields	Medium
TS_04	Delete account and verify the confirmation message	High
Login & Logout		
TS_05	Login with valid email and password	High
TS_06	Login with incorrect email or password	High
TS_07	Login with empty fields	Medium
TS_08	Logout and verify redirect to the login page	High
Products & Search		
TS_09	Verify the all-products list is displayed	High
TS_10	View product details (name, category, price, availability, brand)	High
TS_11	Search for an existing product	High
TS_12	Search for a non-existing product	Medium
TS_13	Filter products by category and by brand	Medium

ID	Scenario	Priority
TS_14	Add a review on a product	Low
Shopping Cart		
TS_15	Add a single product to the cart	High
TS_16	Add multiple products / a specific quantity to the cart	High
TS_17	Verify price, quantity and total in the cart	High
TS_18	Remove a product from the cart	High
Checkout & Payment		
TS_19	Place an order as a new user (register during checkout)	High
TS_20	Place an order as a logged-in user	High
TS_21	Verify delivery and billing address in checkout	High
TS_22	Payment with valid dummy card details	High
TS_23	Payment with empty or invalid card details	High
TS_24	Download invoice after a successful order	Medium
Other Pages		
TS_25	Submit the Contact Us form with a file upload	Medium
TS_26	Verify subscription on the Home and Cart pages	Low
API Testing (Postman)		
TS_27	GET all products list (200)	High
TS_28	POST to search product with and without a parameter (200 / 400)	Medium
TS_29	POST verify login with valid and invalid details	High
TS_30	Create, update, get and delete a user account via API	High

User Acceptance Testing (UAT)

Area	Scenario
User Account	Test user registration, login, logout, and account deletion.
Product & Cart	Test browsing, searching, adding, updating, and removing products from the cart.
Checkout	Test address entry, payment simulation, and order placement.
Order Management	Verify order confirmation and invoice download.

Regression Testing – Features to be Tested

- Verify that user registration and login still function correctly after system updates.
- Validate that adding and removing items in the cart continues to update totals accurately.
- Ensure quantity updates and price recalculations are accurate.
- Ensure that product images, prices, and availability information are displayed accurately.
- Validate data consistency between UI and database (for user, product, and order records).

8. Execution Strategy

Entry Criteria

Testing starts once the following conditions are met. If any is missing, the QA team assesses the risk and recommends whether to proceed or delay.

Criteria	Owner
Test environment: https://www.automationexercise.com/ is accessible and stable.	Test Team
Test data: dummy users, products, addresses and test card details are prepared.	Test Team
Tools: Selenium WebDriver, TestNG, Postman and Jira are installed and configured.	Test Team
Test documents: test scenarios and test cases are written, reviewed and approved.	Test Team
Core modules: Signup/Login, Products, Cart and Checkout are working.	Test Team
Browsers: Chrome, Firefox and Edge are available.	Test Team

Exit Criteria

Testing is complete when the following targets are reached. If any is not met, the QA team assesses the impact on release quality and recommends follow-up testing.

Criteria	Owner
Test case execution: 100% of planned test cases executed.	Test Team
Pass rate: at least 90%.	Test Team
Critical / High defects: none open.	Test Team
Medium / Low defects: logged and moved to a future release or change request.	Test Team
Defect tracking: all defects logged in Jira.	Test Team
Results documentation: expected and actual results recorded for every test case.	Test Team
Reports: Test Summary and Defect reports approved by the Team Leader.	Team Leader
Environment: cleaned up and backed up.	Test Team

Validation & Defect Management

- It is expected that the testers will execute all the scripts in each of the cycles described above.
- The defects will be tracked using Jira.
- It is the responsibility of the tester to open the defects, retest and close the defect.

9. Timeline & Milestones

Week / Period	Phase / Focus Area	Specific Activities & Tasks
Week 1	Test Planning & Setup	• **Requirement Analysis:** Review user workflows and system specifications for Automation Exercise. • **Environment Verification:** Validate accessibility and stability of the test environment. • **Test Data Setup:** Create dummy user credentials, product items, shipping addresses, and payment data.

Week / Period	Phase / Focus Area	Specific Activities & Tasks
Week 2	Manual Functional Testing	• **User Module:** Execute manual test cases for Registration, Login, Logout, and Account Details. • **Product Module:** Test Product Search, Category & Brand filtering, and Product Details view. • **Shopping Cart Module:** Validate Add/Remove items, Quantity updates, and Coupon applications. • **Defect Management:** Log identified UI and functional defects in Jira with clear steps to reproduce.
Week 3	E2E Checkout & API Testing	• **Checkout Module:** Execute End-to-End Checkout process, Address verification, and Payment simulation. • **Backend API Testing:** Execute Postman collections for Auth, Products, Cart, and Order endpoints. • **Bug Retesting:** Retest resolved defects and execute Sanity checks on new build deployments.
Week 4	Automation & Regression	• **Script Development:** Develop and execute Selenium / Cypress automation scripts for core modules. • **Regression Execution:** Run automated regression suite across Chrome, Firefox, and Microsoft Edge. • **Data Integrity:** Validate database data consistency with UI records (MySQL checks).
Week 5	Integration & UAT Support	• **Compatibility Testing:** Conduct cross-browser and cross-device responsive UI testing. • **UAT Support:** Facilitate User Acceptance Testing scenarios and validate end-to-end user journeys. • **Exploratory Testing:** Perform exploratory and edge-case testing on high-risk features.
Final Week	Test Closure & Reporting	• **Exit Criteria Validation:** Verify 100% test execution, >=90% pass rate, and 0 open Critical/High defects. • **Final Deliverables:** Generate Test Execution Report and Test Summary Report. • **Environment Cleanup:** Perform test data cleanup and archive execution logs and screenshots.

10. Defect Severity

Severity	Impact	Examples
Critical	System crash or core functionality failure — the user can't use the site at all.	**Examples:** • Checkout or payment page not loading. • "Add to Cart" button not working. • Login/Register completely broken. • Server error (500) appears on main pages.
High	Major feature is not working properly but system is still partially usable.	**Examples:** • Product images not showing on the product page. • Coupon codes are not applied during checkout. • Invoice cannot be downloaded after a successful order. • Wrong total price calculation.

Severity	Impact	Examples
Medium	Minor functionality or layout issue that doesn't stop core usage.	**Examples:** • Misaligned buttons or wrong font size. • Product description overlaps slightly. • Category or brand filter returns incomplete results. • Slight delay in loading non-critical pages.
Low	Cosmetic or low-priority issues that have almost no business impact.	**Examples:** • Typo in text (e.g., "Shoppig" instead of "Shopping"). • Small color mismatch from design. • Outdated copyright text in footer. • Missing favicon.

11. Team

Business Area	Testers
E-commerce / Online Store	• Mohamed Alhassan Ali (leader) • Malak Mahmoud Zain • Seif Tarek • Noran Mohammed • Nouran Samir • Salma Osama

12. Project & Testing Dependencies

The testing process depends on the following requirements:

1. **Website Availability:** The Automation Exercise website must be accessible and working properly during testing.
2. **Internet Connection:** A stable internet connection is required to access the website and execute test cases.
3. **Browser Availability:** Google Chrome or Microsoft Edge must be installed and updated.
4. **Automation Tools:** Selenium WebDriver, Java, and TestNG must be installed and configured correctly.
5. **Test Data:** Valid and invalid test data, including user accounts, email addresses, and product information, must be available.
6. **Test Account:** A test user account is required to test login, logout, and checkout functionalities.
7. **Team Coordination:** Team members must share test results and report any issues found during testing.

Contingency Plan

If the website becomes unavailable, the testing team will temporarily stop execution and continue testing when the website becomes accessible again.

13. Risk Management

Risk Item / Description	Severity	Likelihood	Mitigation Strategy	Contingency Plan
Website / Server Downtime The Automation Exercise server might experience downtime, slowness, or instability during execution.	Critical	Medium	Regularly monitor environment status and configure explicit waits/timeouts in automation scripts.	Re-run failed test cases at another time or run tests locally if access is available.

Risk Item / Description	Severity	Likelihood	Mitigation Strategy	Contingency Plan
UI Elements & Dynamic Locators Change Automation scripts may fail due to locator changes, dynamic IDs, or newly introduced pop-up ads.	Medium	High	Utilize robust locator strategies (such as data-qa attributes or flexible XPaths) and keep scripts updated.	Temporarily convert affected automated tests into manual test cases until locator updates are complete.
Test Data Duplication / Corruption User registration scenarios might fail because accounts or emails already exist in the database.	Medium	High	Generate dynamic test data (e.g., using timestamps or appending unique UUIDs to email addresses).	Execute database cleanup scripts or reset test data before test execution cycles.
Checkout & Payment Simulation Failures Users are unable to complete purchases, or prices, taxes, and discounts are calculated incorrectly.	Critical	Medium	Focus intensive manual and automated testing on the E2E flow: Add to Cart -> Checkout -> Payment Simulation -> Order Confirmation.	Log critical defects immediately with top priority for resolution before proceeding to UAT.
Performance & Search Latency Delays or timeouts occur when searching products, applying filters, or browsing large product categories.	Medium	Medium	Validate database query efficiency and optimize backend product API endpoint responses.	Document performance bottlenecks as Medium severity issues and request optimization in future releases.
Tight Schedule & Deliverable Delays Insufficient time to execute complete manual and automated regression test cycles before the target date.	High	Medium	Implement Risk-Based Testing, prioritizing critical user paths, Smoke tests, and core functional scenarios.	Focus automation coverage strictly on high-priority repetitive modules and defer non-critical testing types.

14. Test Deliverables

Deliverable	Contents
Test Plan	Defines the testing objectives, scope, strategy, resources, schedule, and testing activities.
Test Scenarios	Identifies the main application features and user workflows to be tested.
Manual Test Cases	Documents test steps, preconditions, test data, expected results, and execution status for manual testing.
Test Data	Contains the dummy user accounts, product information, addresses, and other data required for testing.
Automation Test Scripts	Contains automated test scripts developed using Selenium WebDriver and Cucumber to verify selected website functionalities.
Automation Execution Report	Summarizes automation execution results, including passed, failed, and skipped scenarios.
API Test Cases	Documents API endpoints, HTTP methods, request data, expected status codes, and response validations.

Sample Test Case

ID	Title	Pre-condition	Steps	Expected	Status
TC_001	Login with valid credentials	A registered user account exists, and the login page is accessible.	1. Open the website 2. Navigate to the Login page. 3. Enter a valid email address. 4. Enter the correct password. 5. Click the Login button.	The user should log in successfully, and the website should display the logged-in account information.	Not Executed

Sample Bug Report

Field	Example
Bug ID	BUG_001
Bug Title	Login error message is not displayed for invalid credentials
Module	Login
Severity	Medium
Priority	Medium
Preconditions	The website is accessible, and the user is on the Login page.
Steps to Reproduce	Steps to Reproduce
Expected Result	An appropriate error message should be displayed, informing the user that the login credentials are incorrect.
Actual Result	To be recorded after executing the test and observing the website's actual behavior.
Status	Open — only if the defect is confirmed
Test Environment	To be recorded during testing.

Field	Example
Evidence	Screenshot or other relevant evidence, if available.